2.4 Pre-Quiz: Sine Rule, Cosine Rule and the Area of a Triangle — Markscheme

Total: 40 marks. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. In Problems 2, 4 and 6 the calculator must be in radian mode: a value worked in degree mode earns the M mark if the method is right, and A0.

1. [Maximum mark: 7]

(a) $\dfrac{\sin B\hat{A}C}{15} = \dfrac{\sin 108^\circ}{21}$ **M1**

$\sin B\hat{A}C = \dfrac{15\sin 108^\circ}{21} = 0.679326\ldots$

$B\hat{A}C = 42.8^\circ \quad (42.7910\ldots)$ **A1**

M1 for a correct sine rule equation pairing each side with its opposite angle, in either orientation. The obtuse value 137.2° *is impossible because the triangle already has an obtuse angle at* B*.*

(b) $A\hat{C}B = 180^\circ - 108^\circ - 42.7910\ldots^\circ = 29.2^\circ \quad (29.2089\ldots)$ **A1**

$\dfrac{AB}{\sin 29.2089\ldots^\circ} = \dfrac{21}{\sin 108^\circ}$ **M1**

$AB = 10.8\,\text{cm} \quad (10.7753\ldots)$ **A1**

Accept the cosine rule: $21^2 = AB^2 + 15^2 - 2(15)(AB)\cos 108^\circ$*, solved on the GDC (the A1 for* $A\hat{C}B$ *is then implied by a correct final answer). FT from (a).*

(c) Area $= \frac{1}{2}(15)(21)\sin 29.2089\ldots^\circ$ **M1**

Area $= 76.9\,\text{cm}^2 \quad (76.8594\ldots)$ **A1**

Accept any two sides with the angle between them, e.g. $\frac{1}{2}(15)(AB)\sin 108^\circ$*. Accept* 76.8 *from the rounded* $A\hat{C}B = 29.2^\circ$*. FT from (a) or (b).*

Feedback: carry the full calculator value of each angle forward; round only the final answer.

2. [Maximum mark: 6]

(a) $QR^2 = 9^2 + 7^2 - 2(9)(7)\cos 1.3$ **M1**

$QR^2 = 96.2951\ldots$

$QR = 9.81\,\text{cm} \quad (9.81300\ldots)$ **A1**

Degree mode gives $QR = 2.01$*: M1A0.*

(b) Area $= \frac{1}{2}(9)(7)\sin 1.3$ **M1**

Area $= 30.4\,\text{cm}^2 \quad (30.3520\ldots)$ **A1**

Degree mode gives 0.715*: M1A0.*

(c) $\dfrac{\sin P\hat{Q}R}{7} = \dfrac{\sin 1.3}{9.81300\ldots}$ **M1**

$\sin P\hat{Q}R = 0.687343\ldots$

$P\hat{Q}R = 0.758 \quad (0.757825\ldots)$ **A1**

Accept the cosine rule, $\cos P\hat{Q}R = \frac{9^2+QR^2-7^2}{2(9)(QR)}$*. The answer must be in radians:* $43.4°$ *earns M1A0.* $P\hat{Q}R$ *is acute because it is opposite the shortest side. FT from (a).*

3. [Maximum mark: 6]

(a) $\cos A\hat{B}C = \dfrac{120^2 + 85^2 - 140^2}{2(120)(85)}$ **M1**

$\cos A\hat{B}C = \dfrac{2025}{20\,400} = 0.0992647\ldots$

$A\hat{B}C = 84.3°$ $\quad (84.3031\ldots)$ **A1**

M1 for the cosine rule with $AC = 140$ as the side opposite $A\hat{B}C$.

(b) Area $= \frac{1}{2}(120)(85)\sin 84.3031\ldots°$ **M1**

Area $= 5070\,\text{m}^2$ $\quad (5074.81\ldots)$ **A1**

Accept 5.07×10^3. Accept any two sides with the angle between them, e.g. $\frac{1}{2}(120)(140)\sin B\hat{A}C$ with $B\hat{A}C = 37.2°$. FT from (a).

(c) $\frac{1}{2}(140)h = 5074.81\ldots$ **M1**

$h = 72.5\,\text{m}$ $\quad (72.4973\ldots)$ **A1**

M1 for recognizing that the shortest route is perpendicular to AC, by any method: the area with base AC, or $h = 120\sin B\hat{A}C$, or $h = 85\sin B\hat{C}A$. Accept 72.4 from the rounded area 5070. FT from (b).

Feedback: carry the full area into (c); round only the final answer.

4. [Maximum mark: 6]

(a) $\dfrac{\sin A\hat{C}B}{10} = \dfrac{\sin 0.6}{7}$ **M1**

$\sin A\hat{C}B = \dfrac{10\sin 0.6}{7} = 0.806632\ldots$

$A\hat{C}B = 0.938 \quad (0.938431\ldots)$ **A1**

$A\hat{C}B = \pi - 0.938431\ldots = 2.20 \quad (2.20316\ldots)$ **A1**

Both values are possible because $0.6 + 2.20316\ldots < \pi$*. Degree mode gives* $\sin A\hat{C}B = 0.0150$*: M1A0A0. Answers in degrees (*$53.8°$*,* $126°$*) earn M1A0A0.*

(b) $A\hat{B}C = \pi - 0.6 - 2.20316\ldots$

$A\hat{B}C = 0.338 \quad (0.338431\ldots)$ **A1**

A bare correct answer earns A1. Accept 0.342 *from the rounded* $A\hat{C}B = 2.20$*. FT from (a).*

(c) Area $= \frac{1}{2}(10)(7)\sin 0.338431\ldots$ **M1**

Area $= 11.6\,\text{cm}^2 \quad (11.6202\ldots)$ **A1**

Accept any two sides with the angle between them, e.g. $\frac{1}{2}(10)(AC)\sin 0.6$ *with* $AC = 4.12$ *from the sine rule. Accept* 11.7 *from the rounded* $A\hat{C}B = 2.20$*. FT from (b).*

Feedback: here rounding $A\hat{C}B$ *to* 2.20 *moves the area from* 11.6 *to* 11.7*. Store the full value in the calculator and use it.*

5. [Maximum mark: 8]

(a) $BD^2 = 6^2 + 9^2 - 2(6)(9)\cos 75°$ **M1**

$BD^2 = 89.0475\ldots$

$BD = 9.44\,\text{cm} \quad (9.43650\ldots)$ **A1**

(b) $\dfrac{\sin B\hat{D}C}{8} = \dfrac{\sin 62°}{9.43650\ldots}$ **M1**

$\sin B\hat{D}C = 0.748538\ldots$

$B\hat{D}C = 48.5° \quad (48.4639\ldots)$ **A1**

The obtuse value $131.5°$ *is impossible:* $131.5° + 62° > 180°$. *Accept* $48.4°$ *from the rounded* $BD = 9.44$. *FT from (a).*

(c) $C\hat{B}D = 180° - 62° - 48.4639\ldots° = 69.5360\ldots°$ **A1**

Area $ABD = \frac{1}{2}(6)(9)\sin 75° = 26.0799\ldots$

Area $BCD = \frac{1}{2}(8)(9.43650\ldots)\sin 69.5360\ldots° = 35.3639\ldots$ **M1**

Area $ABCD = 26.0799\ldots + 35.3639\ldots$ **M1**

Area $ABCD = 61.4\,\text{cm}^2 \quad (61.4439\ldots)$ **A1**

A1 for $C\hat{B}D$ *(may be seen in the area expression). First M1 for a correct area expression for triangle* BCD *with two sides and the angle between them, by any route, e.g.* $\frac{1}{2}(BD)(CD)\sin B\hat{D}C$ *with* $CD = 10.0$ *from the sine rule. Second M1 for adding the areas of the two triangles. Accept* 61.5 *from the rounded* $BD = 9.44$ *and* $B\hat{D}C = 48.4°$. *FT from (a) and (b).*

Feedback: carry BD *and* $B\hat{D}C$ *forward at full precision.*

6. [Maximum mark: 7]

(a) Attempt to use the area formula with x, $x+4$ and 2.1 **(M1)**

$\frac{1}{2}x(x+4)\sin 2.1 = 30$ **A1**

$x = 6.57 \quad (6.57368\ldots)$ **A1**

Accept the equivalent quadratic $x^2+4x-69.5080\ldots=0$. *The root* $x=-10.6$ *is rejected because a length is positive; giving both roots as the final answer earns A0. Degree mode gives* $x=38.5$*: A0.*

(b) $BC^2 = 6.57368\ldots^2 + 10.5736\ldots^2 - 2(6.57368\ldots)(10.5736\ldots)\cos 2.1$ **M1**

$BC^2 = 225.197\ldots$

$BC = 15.0\,\text{cm} \quad (15.0065\ldots)$ **A1**

FT from (a).

(c) $\dfrac{\sin A\hat{B}C}{10.5736\ldots} = \dfrac{\sin 2.1}{15.0065\ldots}$ **M1**

$\sin A\hat{B}C = 0.608219\ldots$

$A\hat{B}C = 0.654 \quad (0.653815\ldots)$ **A1**

M1 for a sine rule or cosine rule equation with $AC = x+4$ *as the side opposite* $A\hat{B}C$. $A\hat{B}C$ *is acute because* $B\hat{A}C$ *is obtuse. The answer must be in radians:* $37.5°$ *earns M1A0. FT from (a) and (b).*